WORLD CUP PREDICTION MODEL

Sources & Provenance

Every external input used by the World Cup prediction project, what it contributed, and where its limits begin.

SCOPE	STATUS	PREPARED
2026 World Cup model v2.14	**Research prototype**	**13 October 2026**

PRINCIPLE	**The model was assembled from traceable public/open data, live provider snapshots, and explicit manual overrides. Reference material was never presented as a hidden proprietary feed.**

1. Foundation data

These sources built the historical, tournament, player, tactical, and country-level foundation. Counts below describe the local processed snapshot, not a promise of permanent upstream availability.

Source	Local snapshot	How it entered the model	Boundary
International results Mart Jurisoo; CC0 on Kaggle	49,296 scored senior internationals, shootouts, scorers and former team names.	Built local Elo history, recent form, time-safe training rows and shootout support.	Local Elo is a project calculation, not an official rating.
StatsBomb Open Data Open event data; attribution required	2018 and 2022 World Cup metadata, lineups and event files: 128 matches / 256 team-match summaries.	Prototyped shots, xG, pressure, passing, duels, dribbles and set-piece profiles.	Open subset is not a live commercial feed.
OpenFootball World Cup Public fixture project	2026 schedule structure: 104 fixtures and 48 teams.	Group simulation, bracket routing, rest-day and venue joins.	Cross-checked against official tournament updates.
Public World Cup squad tables Public compilation citing official squads	1,246 players plus names, positions, clubs, caps and goals; 2018/2022 snapshots for historical tests.	Seeded the custom Player Impact Rating and expected-XI / bench depth layers.	Ratings are model-derived, not EA/FIFA/Transfermarkt scores.
World Bank WDI CC BY 4.0 unless marked otherwise	GDP per capita and population snapshots.	Low-weight Klement-style country context and research comparisons.	Not allowed to dominate football strength.
FIFA Men's World Ranking FIFA terms	Saved ranking-page reference snapshot.	External sanity check and optional feature research.	Use and redistribution require care under FIFA terms.

2. Live and operational sources

These feeds moved the project from a static tournament forecast to repeatable matchday runs. Every prediction depended on what was actually available before kickoff; unavailable lineup or weather rows stayed marked as missing.

Source	What it supplied	Operational treatment
API-Football / API-Sports Authenticated API	Fixtures, kickoff/status/results, venues, lineups, events and statistics; odds or availability where returned.	Cached calls, a local request ledger and a reserve protected the plan limit. Provider gaps were never silently filled.
U.S. National Weather Service Public U.S. forecast API	Hourly forecast for U.S. venues.	Primary U.S. source for kickoff temperature, wind and precipitation context.
Open-Meteo Public weather API	Hourly venue forecast worldwide.	Primary outside the U.S. and fallback inside it; used as a bounded tempo / goals adjustment.
FIFA official schedule/results Official tournament source	Schedule and result verification.	Cross-check for provider IDs, knockout routing and final scoring.
Oddschecker public outright snapshot Public article snapshot	Top-team World Cup winner prices published 3 June 2026.	A 5% market sanity layer in v2.3, never copied as the answer.
Road to 26 Public fixture listing	Practice / friendly fixture discovery before the tournament.	Used to exercise v2.4 provider and weather plumbing.
Confirmed team sheets and user evidence Official/public pages or user-supplied screenshots	Authoritative starting XIs when the API returned none or the user supplied the exact lineup.	Stored as a manual overlay with provenance; never used to rewrite the historical base silently.
ClubElo (supplemental) Public club-rating reference	Club strength for post-tournament MLS and club-match portability checks.	Not a World Cup training input and not evidence that the club model is production-ready.

3. Research and benchmark sources

Reference	Role
Panmure Liberum / Joachim Klement note Primary public description	Inspired the question: can macro context, ranking strength and randomness help forecast a World Cup? Exact coefficients and raw model data were not public.
BBC Sport context Secondary reporting	Added context on Klement's record, uncertainty and the role of luck; treated as journalism, not training data.
Hoffmann, Ging & Ramasamy (2002) Academic background	Economic, population and climate variables in national-team performance; used to understand the lineage of the econometric idea.
World Bank Climate Knowledge Portal Climate-data research source	Researched for the Klement-style average-temperature concept. Matchday weather ultimately came from NWS/Open-Meteo instead.

4. How the data became a prediction

1. Acquire and freeze. Public files and provider responses were saved with source notes, dates and identifiers whenever terms allowed.
2. Normalize. Team names, match dates, venues, player identities and competition states were standardized into processed tables.
3. Build only past-facing features. Elo, form and training features were calculated from information available before the prediction timestamp.
4. Score and calibrate. Candidate models were compared on chronological holdouts; the winning ensemble was calibrated before public gating.
5. Apply matchday context. Weather, confirmed lineups and replacement gaps could move probabilities within explicit caps.
6. Freeze, publish and validate. Pre-kickoff calls were archived, then scored later. Post-kickoff backfills were excluded from honest accuracy totals.

5. Retention and reproducibility

- Processed datasets, model artifacts, API/weather caches, validation ledgers, reports and reproducible scripts were retained.
- Regenerable StatsBomb event and 360 JSON payloads were removed during the 22 June storage cleanup, recovering about 837 MiB. The downloader remains in scripts/download_statsbomb_worldcup.py.
- Secrets are excluded from shared ZIP files. A recipient supplies their own API key through the local environment file.

Scope note: later MLS, PSG-Aston Villa and Nations League exercises tested portability with ClubElo or public team sheets. They did not retrain the World Cup model and remain separate experiments.

6. Commercial and scientific limits

Area	Honest position
Licensed data	No Opta/Stats Perform, Sportradar, Wyscout, SkillCorner or paid StatsBomb live feed was acquired. Open data is excellent for research, but licensing must be solved before resale.
Odds	The project used a dated public outright snapshot and occasional provider odds. It did not prove an edge against no-vig closing prices.
Lineups	Provider coverage varied. Manual lineups were accepted only when explicitly supplied as authoritative and remained auditable.
Accuracy	The pre-final World Cup record is evidence from one tournament, not a guaranteed future hit rate. Confidence intervals remain wide.

Bottom line: the source trail is sufficient to reproduce and audit the research model. A sellable product still needs licensed live data, automated immutable snapshots, and independent benchmark testing.